# Even-Block Gap Closure at $k = 4$ for the Rank Isolation Lemma

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**Abstract**

We prove that the even-block gap in the Rank Isolation Lemma closes unconditionally at $k = 4$. The Rank Isolation Lemma asserts that rank drops in the staircase symmetrizing product $\Pi = \prod(1 + s_{i_j})/2$ for the longest element $w_0 \in \mathfrak{S}_n$ occur *only* at first appearances of the $H$-generators $s_1, s_3, s_5, \ldots$, where $H = \langle s_1, s_3, s_5, \ldots\rangle$. The within-block cascade and odd block starts are proved elsewhere. The even-block gap—whether $P_k$ preserves rank at the start of an even block—reduces by cascade transport to a single step within each block, and by parabolic reduction to $\mathfrak{S}_{k+1}$ irreps.

We show that in any $\mathfrak{S}_n$-representation ($n \geq 5$), the intersection $E_1^{(+1)} \cap E_3^{(+1)} \cap \ker\big((1+s_4)(1+s_2)/4\big)$ is trivial. The proof uses two applications of an *adjacent disjointness* lemma for eigenspaces of consecutive Coxeter generators. We explain why the analogous argument fails for $k \geq 6$. The result has been verified computationally for all irreps of $\mathfrak{S}_n$, $n = 5, 6, 7$.

## 1 Context and setup

### 1.1 The staircase product

Let $\mathfrak{S}_n$ act on a finite-dimensional complex vector space $V$ via simple transpositions $s_1, \ldots, s_{n-1}$ satisfying the usual Coxeter relations. Define the *staircase word* for the longest element $w_0$ to be

$$s_1,\ s_2 s_1,\ s_3 s_2 s_1,\ \ldots,\ s_{n-1} s_{n-2} \cdots s_1,$$

and the *staircase product*

$$\Pi \;=\; \prod_j \frac{1 + s_{i_j}}{2},$$

where the factors are taken in the order prescribed by the staircase word (left to right). Each factor $P_i = (1 + s_i)/2$ is an idempotent projection onto the $+1$ eigenspace $E_i^{(+1)}$ of $s_i$.

### 1.2 The $H$-invariant subspace

Let $H = \langle s_1, s_3, s_5, \ldots\rangle \leq \mathfrak{S}_n$ be the subgroup generated by odd-indexed simple transpositions. The *Rank Isolation Lemma* (conjectured and verified computationally through $n = 16$) states:

[Rank Isolation] For any irreducible $\mathfrak{S}_n$-representation $V_\lambda$,

$$\operatorname{rank}\big(\Pi\big|_{V_\lambda}\big) \;=\; \dim V_\lambda^H.$$

Moreover, the rank drops occur *only* at the first appearance of each $H$-generator $s_{2m-1}$ in the staircase word.

Three of the four injectivity components are proved unconditionally:

1. **Within-block cascade:** Adjacent disjointness gives $E_i^{(-1)} \cap E_{i+1}^{(+1)} = \{0\}$, making each step within a block injective.

2. **Block 2 start:** $P_2$ on $E_1^{(+1)}$ is injective by adjacent disjointness.

3. **Odd block starts:** For $s_k \in H$ (odd $k$), the transported image already lies in $E_k^{(+1)}$, and the contraction argument proves injectivity.

The remaining gap is at *even* block starts: when block $k$ begins with $P_k$ and $k$ is even, $k \geq 4$.

### 1.3 Reduction of the even-block gap

Two reductions narrow the gap:

- **Cascade transport** (proved separately): The gap at the start of even block $k$ reduces to a *single step* within block $k-1$, namely $P_{k-2}$ applied immediately after $P_{k-1}$.

- **Parabolic reduction** (proved separately): Even block $k$ holds for all $n$ if and only if it holds for all $\mathfrak{S}_{k+1}$ irreps.

Thus, closing even block $k = 4$ requires proving a statement about $\mathfrak{S}_5$ representations—which we do below for arbitrary $\mathfrak{S}_n$.

## 2 Adjacent disjointness

The following elementary lemma is the engine of the proof.

**Lemma 2.1** (Adjacent disjointness)**.** *Let $V$ be any representation of $\mathfrak{S}_n$ ($n \geq 3$), and let $s_i, s_{i+1}$ be consecutive simple transpositions. Then*

$$E_i^{(+1)} \cap E_{i+1}^{(-1)} \;=\; \{0\} \qquad \textit{and} \qquad E_i^{(-1)} \cap E_{i+1}^{(+1)} \;=\; \{0\},$$

*where $E_j^{(\pm 1)} = \{v \in V : s_j\, v = \pm v\}$.*

*Proof.* The subgroup $\langle s_i, s_{i+1} \rangle \cong \mathfrak{S}_3$ acts on $V$ by restriction. It suffices to verify the claim on each irreducible $\mathfrak{S}_3$-representation.

- **Trivial representation** (dim = 1): $s_i$ and $s_{i+1}$ both act as $+1$. So $E_i^{(+1)} = V$, $E_{i+1}^{(-1)} = \{0\}$, and $E_i^{(-1)} = \{0\}$, $E_{i+1}^{(+1)} = V$. Both intersections are trivial.

- **Sign representation** (dim = 1): $s_i$ and $s_{i+1}$ both act as $-1$. So $E_i^{(+1)} = \{0\}$ and $E_{i+1}^{(+1)} = \{0\}$. Both intersections are trivial.

- **Standard representation** (dim = 2): In a suitable basis, $s_i$ and $s_{i+1}$ are represented by reflection matrices with eigenvalues $\{+1, -1\}$. Each eigenspace is one-dimensional. The two reflections correspond to distinct hyperplanes in $\mathbb{R}^2$ (the angle between them is $\pi/3$, since $(s_i s_{i+1})^3 = 1$). In particular, $E_i^{(+1)} \neq E_{i+1}^{(-1)}$ and $E_i^{(-1)} \neq E_{i+1}^{(+1)}$: a $+1$-eigenline of one reflection is not the $-1$-eigenline of the other. Since both eigenspaces are one-dimensional, distinct lines in $\mathbb{R}^2$ meet only at the origin.

Since every $\mathfrak{S}_3$-representation decomposes into these three irreducibles, the result follows by taking direct sums. $\square$

## 3 Even-block closure at $k = 4$

**Lemma 3.1** (Even-block closure at $k = 4$)**.** *Let $V$ be any representation of $\mathfrak{S}_n$ ($n \geq 5$). For any $v \in V$ satisfying $s_1\, v = v$ and $s_3\, v = v$, if*

$$(1 + s_4)(1 + s_2)\, v \;=\; 0,$$

*then $v = 0$.*
*Equivalently:*

$$E_1^{(+1)} \;\cap\; E_3^{(+1)} \;\cap\; \ker\!\left(\frac{(1 + s_4)(1 + s_2)}{4}\right) \;=\; \{0\}.$$

*Proof.* Suppose $(1 + s_4)(1 + s_2)\, v = 0$ with $s_1\, v = v$ and $s_3\, v = v$. Since $s_4$ is an involution, decompose

$$v \;=\; w_+ + w_-, \qquad s_4\, w_+ = w_+, \quad s_4\, w_- = -w_-.$$

**Step 1 (Kill $w_+$).** Since $|1 - 4| = 3 \geq 2$, the transposition $s_1$ commutes with $s_4$. Therefore the $E_4^{(\pm 1)}$ decomposition is preserved by $s_1$, and in particular $s_1\, w_+ = w_+$ (projecting $s_1\, v = v$ to $E_4^{(+1)}$).

Similarly, $|2 - 4| = 2 \geq 2$, so $s_2$ commutes with $s_4$. Thus $(1 + s_2)$ preserves the $E_4^{(\pm 1)}$ decomposition. Using the commutativity $[s_2, s_4] = 0$, we rewrite the hypothesis:

$$0 \;=\; (1 + s_4)(1 + s_2)\, v \;=\; (1 + s_2)(1 + s_4)\, v \;=\; (1 + s_2)(2w_+).$$

Hence $(1 + s_2)\, w_+ = 0$, which gives $s_2\, w_+ = -w_+$, i.e., $w_+ \in E_2^{(-1)}$.

Combining: $w_+ \in E_1^{(+1)} \cap E_2^{(-1)}$. By adjacent disjointness (Lemma 2.1 applied to $s_1, s_2$), this intersection is trivial. Therefore $\boldsymbol{w_+ = 0}$.

**Step 2 (Kill $w_-$).** Since $w_+ = 0$, we have $v = w_- \in E_4^{(-1)}$. But by hypothesis $s_3\, v = v$, so $v \in E_3^{(+1)}$. Hence $v \in E_3^{(+1)} \cap E_4^{(-1)}$. By adjacent disjointness (Lemma 2.1 applied to $s_3, s_4$), this intersection is trivial. Therefore $\boldsymbol{v = 0}$. □

**Corollary 3.2** (Even block $k = 4$ is safe)**.** *At the gap step for even block 4 in the staircase product, $P_2$ preserves the rank of the transported $V^H$ image. Consequently, no rank drop occurs at even block 4.*

*Proof.* By cascade transport, the gap at block 4 reduces to a single step: $P_2$ applied to vectors in the transported $V^H$ image after $P_3$. The transported image at this point satisfies two eigenspace conditions:

1. $s_3\, w = w$: since $P_3 = (1 + s_3)/2$ projects onto $E_3^{(+1)}$.
2. $s_1\, w = w$: the original $V^H$ satisfies $s_1\, v = v$; all subsequent projections in blocks 1 and 2 end with $P_1$, restoring $E_1^{(+1)}$; and $P_3$ uses $s_3$ which commutes with $s_1$ ($|1 - 3| = 2$), preserving $E_1^{(+1)}$.

Suppose $P_2$ annihilates the $E_4^{(+1)}$ component of some nonzero $w$ in this image. Decompose $w = w_+ + w_-$ in $E_4^{(\pm 1)}$. If $(1+s_2)\, w_+ = 0$, then since $s_2$ commutes with $s_4$, we obtain $(1+s_4)(1+s_2)\, w = 0$. But $w \in E_1^{(+1)} \cap E_3^{(+1)}$, so Lemma 3.1 forces $w = 0$—a contradiction.

Therefore $P_2$ is injective on the transported $V^H$ image, and no rank drop occurs. □

## 4 Obstruction for $k \geq 6$

*Remark* 4.1 (Why the argument does not extend to $k = 6$). For even block $k = 6$, the analogous contraction would require showing that

$$E_1^{(+1)} \cap E_5^{(+1)} \cap \ker\left(\frac{(1+s_6)(1+s_4)}{4}\right) = \{0\}.$$

The proof of Lemma 3.1 used two adjacent disjointness applications: first $(s_1, s_2)$ to kill the $E_4^{(+1)}$ component, then $(s_3, s_4)$ to kill the remainder.

At $k = 6$, Step 1 would decompose in $E_6^{(\pm 1)}$ eigenspaces. Since $s_4$ commutes with $s_6$ ($|4-6| = 2$), projecting $(1 + s_6)(1 + s_4)\, v = 0$ to $E_6^{(+1)}$ gives $(1 + s_4)\, v_+ = 0$, hence $v_+ \in E_4^{(-1)}$. Combined with $s_1\, v_+ = v_+$, we get $v_+ \in E_1^{(+1)} \cap E_4^{(-1)}$. But $s_1$ and $s_4$ are *not* adjacent ($|1 - 4| = 3$), so adjacent disjointness does not apply. Indeed, $E_1^{(+1)} \cap E_4^{(-1)} \neq \{0\}$ in general (e.g., in the standard representation of $\mathfrak{S}_7$).

The natural repair would be to use $s_3$ (adjacent to $s_4$) instead of $s_1$. This would require $v \in E_3^{(+1)}$ at the gap step. However, the cascade through blocks 3 and 4 includes $P_4$, which uses $s_4$ adjacent to $s_3$ ($|3 - 4| = 1$), and this breaks the $E_3^{(+1)}$ eigenspace condition. The transported $V^H$ image at the $k = 6$ gap step lies in $E_5^{(+1)} \cap E_1^{(+1)}$ but *not* in $E_3^{(+1)}$.

Computational verification confirms that the gap is safe through $\mathfrak{S}_8$, but a structural proof for $k \geq 6$ remains open.

## 5 Computational verification

All claims have been verified computationally using exact arithmetic over $\mathbb{Q}$ (rational arithmetic in Python/SageMath) via Young's seminormal form.

**Proposition 5.1** (Computational verification)**.** *The following have been verified:*

1. ***Lemma 3.1:*** $E_1^{(+1)} \cap E_3^{(+1)} \cap \ker(Q') = \{0\}$, *where* $Q' = (1 + s_4)(1 + s_2)/4$, *holds for all irreducible representations* $V_\lambda$ *of* $\mathfrak{S}_n$ *with* $n = 5, 6, 7$.

2. $V^H$ ***version:*** $V_\lambda^H \cap \ker(Q') = \{0\}$ *holds for all irreps of* $\mathfrak{S}_n$, $n = 5, 6, 7, 8$.

3. ***Transported*** $V^H$***:*** *The staircase-transported image at the gap step intersects* $\ker(Q')$ *trivially for all irreps through* $\mathfrak{S}_7$.

By parabolic reduction, even block $k = 4$ depends only on $\mathfrak{S}_5$ irreps. Since the algebraic proof of Lemma 3.1 applies to *any* $\mathfrak{S}_n$-representation, the result is unconditional.

## 6 Summary of proof status

| **Component** | **Status** | **Scope** |
|---|---|---|
| Within-block cascade | **Proved** | All $n$ |
| Block 2 start | **Proved** | All $n$ |
| Odd block starts | **Proved** | All $n$ |
| Even block $k = 4$ | **Proved** | All $n \geq 5$ |
| Even block $k = 6$ | Verified | $n \leq 8$ |
| Even block $k = 8$ | Verified | $n \leq 16$ |
| Even block $k \geq 6$ general | **[Gap]** | |

With the $k = 4$ closure, the Rank Isolation Lemma is proved unconditionally for all $n \leq 6$ (blocks $\leq 5$). For $n \leq 16$, the remaining even-block gaps ($k = 6, 8, \ldots, 14$) are closed by computational verification. A uniform proof for all even $k$ remains the principal open problem.